

Connect objects to their nets and build objects from their nets using spatial and geometric reasoning

Learning intention

- connect objects to their nets and build objects from their nets using spatial and geometric reasoning.

Teach and model

- designing and constructing exact nets for packaging particular shaped items or collections of interest.
- sketching nets for a range of prisms and pyramids considering the number, shape and placement of the faces, and test by cutting and folding.
- visualising folding some possible nets for a range of prisms and pyramids, predicating which will work and which cannot work, and justifying their choices.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.